

Hertzsprung–Russell diagram

A plot of **Luminosity (Absolute Magnitude)** against the **color** of the stars ranging from the **high-temperature blue-white stars** to the **low temperature red stars**.

Developed in the early **1900s** by **Ejnar Hertzsprung** and **Henry Norris Russell**.

This a tool used for **study of evolution** of Stars.

Code Explanation

1.Importing all the libraries:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

Seaborn is built on matplotlib & it gives visually better and simpler graphs.

2.Reading the data:

```
fpath = "Dataset.csv"
data = pd.read_csv(fpath)
```

3.Renaming columns in easier names for use case:

```
# Renaming the columns
data.columns = [
    "Temp_K",
    "Luminosity_LLo",
    "Radius_RRo",
    "Absolute_Magnitude_Mv",
    "Star_Type",
    "Star_Color",
    "Spectral_Class"
]
```

4. Converting to numeric values:

```
# Convert to numeric values(i.e.'float' or 'int')
numeric_cols = ["Temp_K", "Luminosity_LLo", "Absolute_Magnitude_Mv"]
data[numeric_cols] = data[numeric_cols].apply(pd.to_numeric, errors="coerce")
```

1. `numeric_cols = [...]`

This creates a list of the column names we want to convert to numeric (numbers).

2. `data[numeric_cols]`

This selects the specific columns from the DataFrame 'data'.

gives a smaller DataFrame containing the specific columns.

3. `pd.to_numeric`

Try to convert all the values in this column into numbers (integers or floats).

4. `errors="coerce"`

Replaces any invalid data with 'NaN' (missing values).

5. Dropping Rows with missing values:

```
# Dropping rows with missing values(i.e. skipping them)
data.dropna(subset=numeric_cols, inplace=True)
print(data)
```

Removes/Passes the rows that have missing values so that the plot doesn't break.

- `inplace=True` → modifies the original DataFrame **directly** (no copy).
- `inplace=False` (default) → returns a *new* DataFrame, leaving the original unchanged.

6. Plotting the Dataset values:

This plots the **HR Diagram** after providing all the specifications required for the graph.

```

# Plot HR Diagram
plt.figure(figsize=(10, 6))
sns.scatterplot(
    x="Temp_K",
    y="Luminosity_LLo",
    hue="Star_Type",
    palette="viridis",          # For Smooth Colour Gradient
    data=data,
    s=100,                     # Dot Size
    alpha=0.8                  # Opacity
)

# Invert x-axis (hotter → cooler)
plt.gca().invert_xaxis()

# Titles and labels
plt.title("Hertzsprung-Russell Diagram", fontsize=14, fontweight='bold')
plt.xlabel("Temperature (K) [Hotter → Cooler]")
plt.ylabel("Luminosity (L/Lo)")
plt.legend(title="Star Type")
plt.grid(True, linestyle="--", alpha=0.5)

# Plotting
plt.show()

```

Explanation About Star Types

Star Type	Stellar Category	Description
0	Brown Dwarf	Very cool, small, low-luminosity objects, below main sequence stars
1	Red Dwarf	Small, cool, dim main-sequence stars (common in galaxy)
2	White Dwarf	Hot but tiny stars (dead cores of old stars)
3	Main Sequence	Normal, hydrogen-burning stars (like our Sun)
4	Supergiant	Extremely bright and large stars
5	Hypergiant	Very massive, extremely luminous stars nearing instability

Data About Some Famous Stars

Star Name	Temperature (K)	Luminosity (L/L _o)	Category (Type)	Description
Proxima Centauri	3042	0.0017	1 – Red Dwarf	Small, cool, and faint star; closest to the Sun.
Our Sun	5778	1.0	3 – Main Sequence	Medium-sized hydrogen-burning star; stable and yellow.
Alpha Centauri A	5790	1.52	3 – Main Sequence	Similar to the Sun but slightly hotter and brighter.
Alpha Centauri B	5260	0.50	3 – Main Sequence	Cooler and dimmer companion of Alpha Centauri A.